
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Asaba, Joshua
Date of birth:	2/24/1990
Subject:	020-AGK
Result:	47,49 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	3654		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
2	3662		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
3	4495		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
4	2722		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
5	2408	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
6	2419	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
7	2397	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
8	2417	21.2.3.3	Loads, stresses flutter	0,00	1,00
9	1271	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	1,00	1,00
10	546	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
11	585	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
12	491	21.8.2.2	Design, operation, system components, etc	0,00	1,00
13	809	21.8.2.2	Design, operation, system components, etc	0,00	1,00
14	1005	21.8.2.2	Design, operation, system components, etc	0,00	1,00
15	1043	21.8.2.2	Design, operation, system components, etc	1,00	1,00
16	926	21.9.1.6	Electromagnetism	1,00	1,00
17	159	21.9.3	Generation	0,00	1,00
18	817	21.9.3.1	DC Generation	0,00	1,00
19	436	21.9.3.2	AC Generation	0,00	1,00
20	1197	21.11.2.2	Compressor and diffuser	1,00	1,00
21	2809	22.	INSTRUMENTATION	1,00	1,00
22	3871	22.	INSTRUMENTATION	1,00	1,00
23	2822	22.	INSTRUMENTATION	1,00	1,00
24	3405	22.	INSTRUMENTATION	1,00	1,00
25	4463	22.	INSTRUMENTATION	1,00	1,00
26	2793	22.	INSTRUMENTATION	1,00	1,00
27	2815	22.	INSTRUMENTATION	1,00	1,00
28	3094	22.	INSTRUMENTATION	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	2819	22.	INSTRUMENTATION	1,00	1,00
30	3112	22.	INSTRUMENTATION	0,00	1,00
31	3014	22.	INSTRUMENTATION	1,00	1,00
32	3107	22.	INSTRUMENTATION	0,00	1,00
33	34	22.	INSTRUMENTATION	1,00	1,00
34	1123	22.2	MEASUREMENT OF AIR DATA PARAMETERS	0,00	1,00
35	17	22.2	MEASUREMENT OF AIR DATA PARAMETERS	1,00	1,00
36	1110	22.2	MEASUREMENT OF AIR DATA PARAMETERS	0,00	1,00
37	1004	22.2.1	Pressure measurement	1,00	1,00
38	242	22.2.4	Altimeter	0,00	1,00
39	107	22.2.6	Airspeed indicator (ASI)	0,00	1,00
40	1144	22.2.6	Airspeed indicator (ASI)	1,00	1,00
Total:48%				19,00	40,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Asaba, Joshua

Location: Uganda Civil Aviation Authority

Your Result: 19,00 from 40,00 Points (47,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	5,00	20,00	25,00
21.2. AIRFRAME	4	1,00	4,00	25,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	1	1,00	1,00	100,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	2	0,00	2,00	0,00
21.8. FUEL SYSTEM	4	1,00	4,00	25,00
21.9. ELECTRICS	4	1,00	4,00	25,00
21.11. TURBINE ENGINES	1	1,00	1,00	100,00
22. INSTRUMENTATION	20	14,00	20,00	70,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	7	3,00	7,00	42,86
Total	40	19,00	40,00	47,50

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Date of birth: 1990-02-24

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1

Many 4-cycle engines utilize what type of lubrication system?

☐ Forced.

☒ Fuel/oil mixture.

☐ Gravity.

2

The fan guard surrounds the propeller and

- ☐ reduces 'P' factor.
- ☒ increases aerodynamic efficiency.
- ☐ protects the parachute suspension lines from damage.

3

What is the CFR Part 1 definition of critical engine?

- ☐ The outboard engine whose failure would most adversely affect safety.
- ☐ The engine whose failure would most adversely affect airplane performance or handling qualities.
- ☒ The engine still running which would most adversely affect performance if it should fail.

4

When refueling aircraft, which precaution would be adequate for eliminating the potential hazard of static electricity?

- ☒ **Ensure that battery and ignition switches are off.**
- ☐ **Connect a ground wire between the aircraft, fuel truck, fuel nozzle, and ground.**
- ☐ **Connect a ground wire from the fuel truck to ground.**

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5

Which direction from the primary control surface does an elevator adjustable trim tab move when the control surface is moved?

- ☐ Same direction.
- ☐ Remains fixed for all positions.
- ☒ Opposite direction.

6

Why must the angle of attack be increased during a turn to maintain altitude?

- ☐ Increase the horizontal component of lift equal to the vertical component.
- ☒ Compensate for loss of vertical component of lift.
- ☐ Compensate for increase in drag.

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7

Which of the following is considered an auxiliary flight control?

- ☒ Upper rudder.
- ☐ Leading-edge flaps.
- ☐ Ruddervator.

8

What affects indicated stall speed?

- ☐ Angle of attack, weight, and air density.
- ☒ Load factor, angle of attack, and power.
- ☐ Weight, load factor, and power.

9

A tubeless tyre has : 1- a built-in-air tube. 2- no built-in-air tube. 3- a crossed side casing. 4- a radial side casing. The combination of correct statements is:

☐ 1, 4.

☐ 1,3.

☐ 2, 3.

☒ 2, 4.

10

The term "Bootstrap", when used to identify a cabin air conditioning and pressurisation system, refers to the :

- ☒ means by which pressurisation is controlled.
- ☐ cold air unit (air cycle machine) arrangement.
- ☐ source of the charge air.
- ☐ charge air across the inter-cooler heat exchanger.

11

In an aircraft air conditioning system the air cannot be treated for:

☐ temperature.

☒ pressure.

☐ purity.

☐ humidity.

12

When leaning the mixture for the most economic cruise fuel flow, excessive leaning will cause :

- ☒ high manifold pressure
- ☐ low cylinder head and exhaust gas temperature
- ☐ high engine rpm
- ☐ high cylinder head and exhaust gas temperature

13

"Vapor lock" is the phenomenon by which:

- ☐ water vapour plugs are formed in the intake fuel line following the condensation of water in fuel tanks which have not been drained for sometime.
- ☐ heat produces vapour plugs in the fuel line.
- ☒ burnt gas plugs forming and remaining in the exhaust manifold following an overheat and thereby disturbing the exhaust.
- ☐ abrupt and abnormal enrichment of the fuel/air mixture following an inappropriate use of carburettor heat.

14

During fuelling the automatic fuelling shut off valves will

- ☐ there is fire.
- ☐ fuelling system has reached a certain pressure.
- ☒ the surge vent tank is filled.
- ☐ the fuel has reached a predetermined volume or mass.

15

The pressurization of tanks is maintained by the fuel:

☐ dump system.

☐ top off unit.

☐ tank drains.

☒ vent system.

16

If a current is passed through a conductor,

- ☐ the current will increase.
- ☐ the intensity of the magnetic field will decrease.
- ☐ there will be no effect unless the conductor is moved.
- ☒ a force will be exerted on the conductor.

17

The output of a generator is controlled by :

- ☐ varying the length of wire in the armature windings.
- ☒ varying the field strength.
- ☐ varying the speed of the engine.
- ☒ the reverse current relay circuit breaker.

18

The reason for using inverters in an electrical system is

- ☐ To change DC into AC.
- ☒ To change AC into DC.
- ☐ To avoid a short circuit.
- ☐ To change the DC voltage.

19

As regards three-phase AC generators, the following conditions must be met for paralleling AC generators: 1. Equal voltage 2. Equal current 3. Equal frequencies 4. Same 5. Voltages of same phase The combination regrouping all the correct statementsphase rotation is :

☐ 1, 3, 4, 5

☒ 1, 3, 5

☐ 1, 2, 3, 4

☐ 1, 4, 5

20

The turbine in a cold air unit (air cycle machine):

- ☐ drives the compressor in the unit and causes a pressure increase in the conditioning air.
- ☐ drives the compressor which provides pressurisation.
- ☒ drives the compressor in the unit, creating a temperature drop in the conditioning air.
- ☐ increases the pressure of the air supply to the cabin.

21

Which instrument would be affected by excessively low pressure in the airplane's vacuum system?

☒ Heading indicator.

☐ Pressure altimeter.

☐ Airspeed indicator.

22

When an altimeter is changed from 30.11" Hg to 29.96" Hg, in which direction will the indicated altitude change and by what value?

- ☐ Altimeter will indicate 15 feet lower.
- ☐ Altimeter will indicate 150 feet higher.
- ☒ Altimeter will indicate 150 feet lower.

23

A possible result of using the emergency alternate source of static pressure inside the cabin of an unpressurized airplane is the

- ☐ **airspeed indicator may indicate less than normal.**
- ☒ **altimeter may indicate an altitude higher than the actual altitude being flown.**
- ☐ **altimeter may indicate an altitude lower than the actual altitude being flown.**

24

How will the airspeed indicator react if the ram air input to the pitot head is blocked by ice, but the drain hole and static port are not?

- ☐ Indication will remain constant but will increase in a climb.
- ☐ Indication will rise to the top of the scale.
- ☒ Indication will drop to zero.

25

(Refer to Figures 139 and 140.) Which displacement from the localizer and glide slope at the 1.9 NM point is indicated?

- ☐ 710 feet to the left of the localizer centerline and 140 feet below the glide slope.
- ☐ 430 feet to the right of the localizer centerline and 28 feet above the glide slope.
- ☒ 710 feet to the right of the localizer centerline and 140 feet above the glide slope.

See Attachments:

- | | | |
|-----------------------|--------------------|-------------------|
| 1. instrument_139.jpg | instrument_139.jpg | (Attachment No.1) |
| 2. instrument_140.jpg | instrument_140.jpg | (Attachment No.2) |

26

In the Northern Hemisphere, if an aircraft is accelerated or decelerated, the magnetic compass will normally indicate

- ☐ a turn momentarily, with changes in airspeed on any heading.
- ☒ correctly when on a north or south heading while either accelerating or decelerating.
- ☐ a turn toward the south while accelerating on a west heading.

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27

If a pitot tube is clogged, which instrument would be affected?

☒ **Airspeed indicator.**

☐ **Altimeter.**

☐ **Vertical speed indicator.**

28

How can a pilot identify a military airport at night?

- ☐ White and red beacon light with dual flash of the white.
- ☒ Green and white beacon light with dual flash of the white.
- ☐ Green, yellow, and white beacon light.

29

If the static pressure tubes are broken inside a pressurized cabin during a high-altitude flight, the altimeter would probably indicate

☒ **lower than actual flight altitude.**

☐ **higher than actual flight altitude.**

☐ **sea level.**

30

What is the advantage of a three-bar VASI?

- ☒ Pilots have a choice of glide angles.
- ☐ The three-bar VASI is much more visible and can be used at a greater height.
- ☐ A normal glide angle is afforded both high and low cockpit aircraft.

31

What is the significance of the blue radial line on the airspeed indicator of a light multiengine airplane and when is it to be used? It indicates the

- ☐ minimum speed at which the airplane is controllable when the critical engine is suddenly made inoperative and should be used at all altitudes when an engine is inoperative.
- ☐ speed which will provide the greatest height for a given distance of forward travel when one engine is inoperative and should be used for all climbs during engine-out operations.
- ☒ speed which will provide the maximum altitude gain in a given time when one engine is inoperative and should be used for climb and final approach during engine-out operations.

32

What are the indications of the pulsating VASI?

- ☐ High - pulsing white, on glidepath - steady white, slightly below glide slope steady red, low - pulsing red.
- ☐ High - pulsing white, on glidepath - green, low - pulsing red.
- ☒ High - pulsing white, on course and on glidepath - steady white, off course but on glidepath - pulsing white and red; low - pulsing red.

33

If the Pitot head is blocked, what airspeed indication can he expected?

- ☐ A decrease of IAS during a climb.
- ☒ No change of IAS in level flight, even with large power changes.
- ☐ Constant IAS during a descent.

34

A VSI may suffer from error due to:

- ☒ compressibility, time lag, blockage
- ☐ pitot blockage, static blockage, aircraft manoeuvre
- ☐ time lag, instrument calibration, blockage
- ☐ density, aircraft manoeuvre, position of the static vent

35

If the Pitot Head and Static Vent were blocked by ice, which instruments would be affected?

- ☐ ASI, Altimeter and Slip indicator.
- ☐ The ASI would under read.
- ☒ Altimeter, VSI and ASI would give inaccurate readings.

36

If the pitot line is blocked, the VSI will:

☒ read zero ft/mm rate of climb/descent

☐ under-read

☒ read normally

☐ over-read

37

A simple pressure altimeter uses:

- ☐ pitot pressure fed to an open capsule
- ☐ static pressure fed outside a completely evacuated aneroid capsule.
- ☒ static pressure fed into a gas tight casing, but outside a partially evacuated capsule
- ☐ static pressure fed to an open capsule

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38

Transition altitude is obtained from:

☐ Jeppesen or Aerad flight guides.

☐ ATC or VOR ATIS.

☒ 1500 feet above airfield elevation.

39

An aircraft is maintaining FL 120 in cloud. The ASI reading falls to zero. The most probable cause is:

☐ Static vent blocked by ice.

☐ ASI malfunction.

☒ Pitot head and static vent blocked by ice.

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40

During a climb, the static line becomes blocked, the ASI will:

- ☐ read zero
- ☐ read normally
- ☐ over-read
- ☒ under-read

Attachment No. 1

Questions 25

instrument_139.jpg

instrument_139.jpg

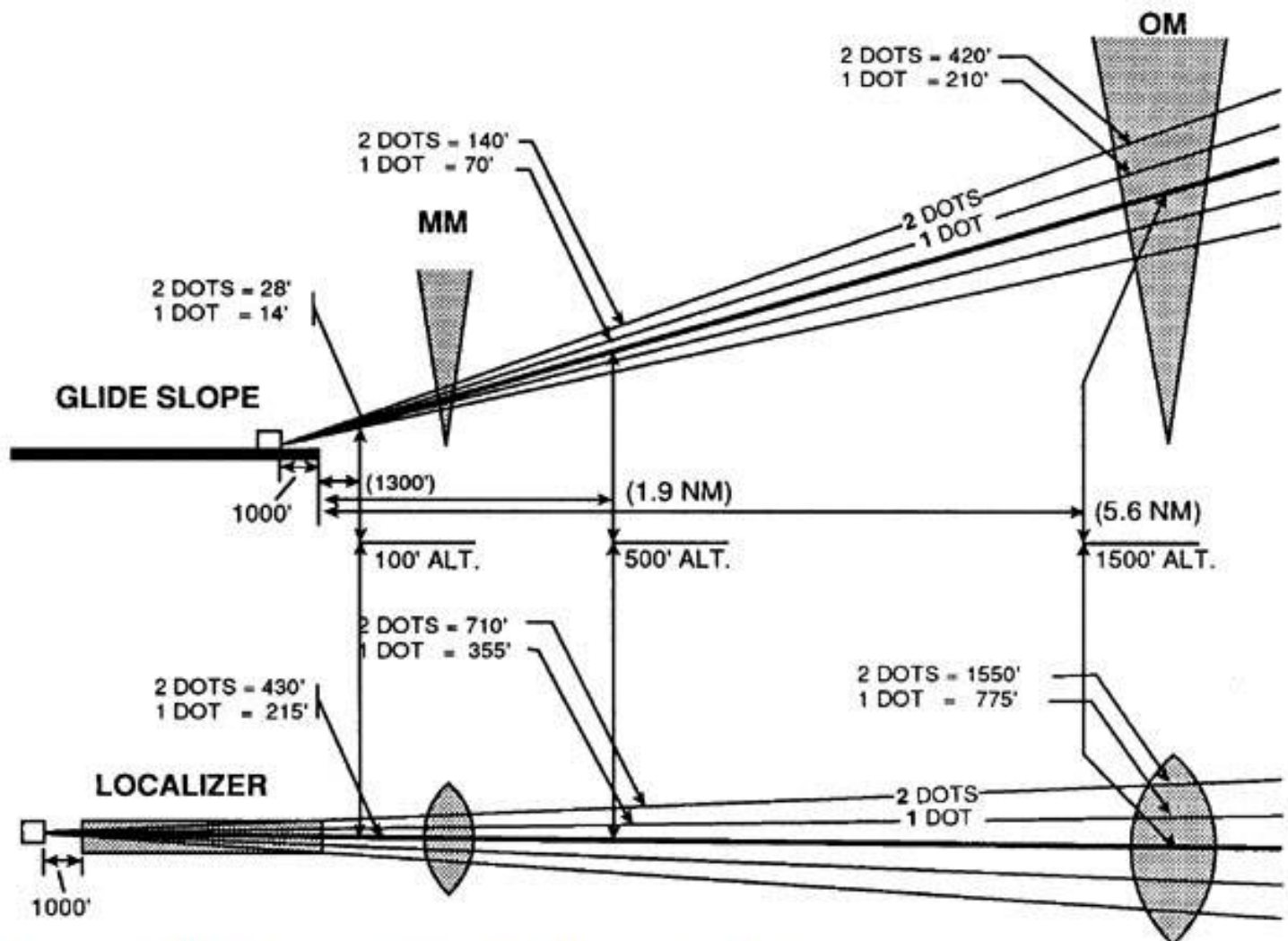


Figure 139. Glide Slope and Localizer Illustration © ASA

Attachment No. 2

Questions 25

instrument_140.jpg

instrument_140.jpg

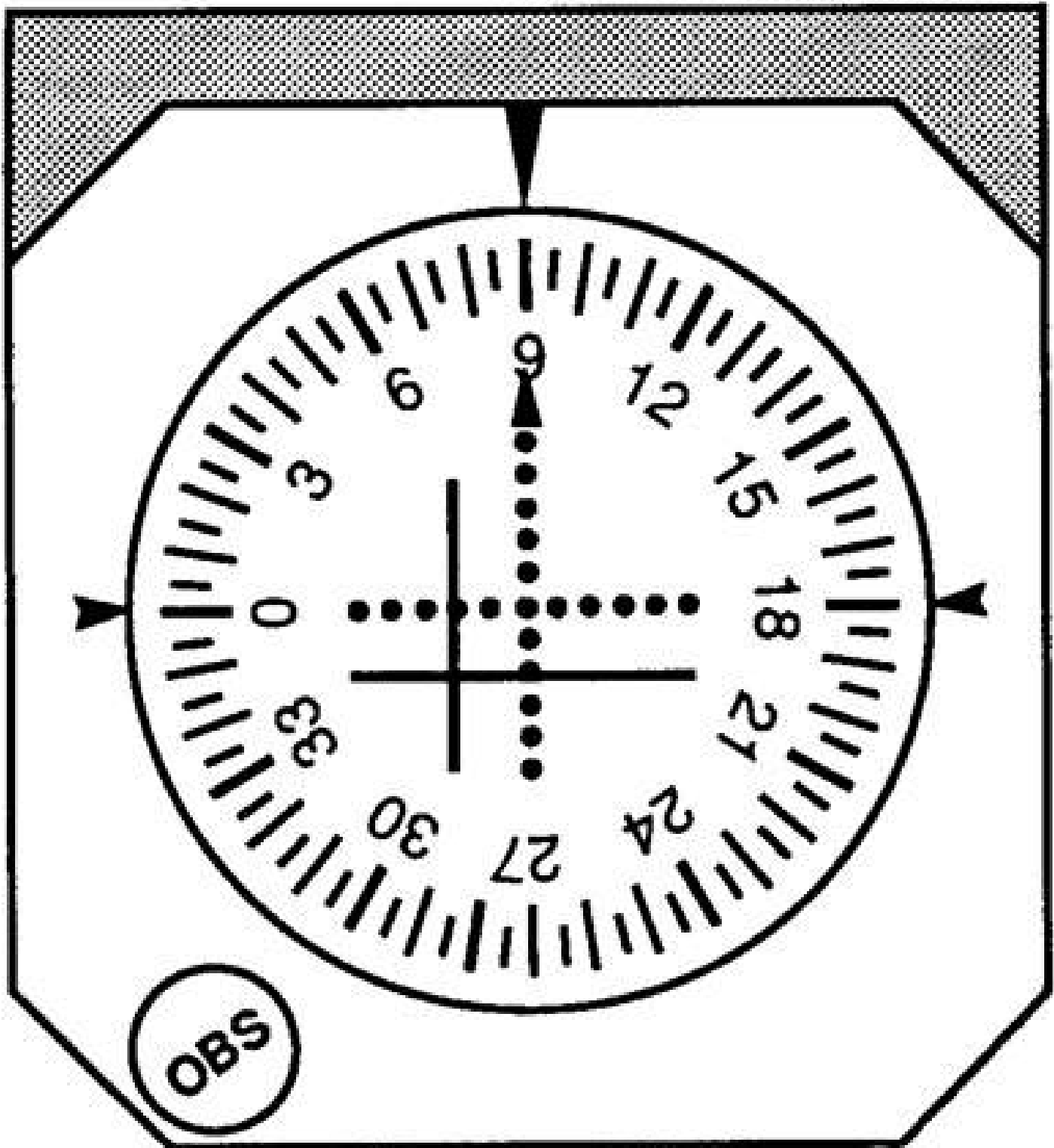


Figure 140. OBS, ILS, and GS Displacement © ASA

